

Ex.No. 2: Implementing TDMA Slot Allocation in GSM.

Date:12-8-26

AIM: To implement TDMA (Time Division Multiple Access) slot allocation in GSM using MATLAB and analyze system performance in terms of time slot duration, frame utilization, and number of allocated slots.

Objective:

To calculate the Time Slot Duration in a GSM TDMA frame by dividing the total frame time into equal time slots. This helps in understanding the time structure of GSM communication.

Procedure:

1. Define GSM frame parameters such as total frame duration and number of time slots.
2. Compute time slot duration by dividing frame duration by number of slots.
3. Assign time slots to multiple users based on availability.
4. Calculate frame utilization percentage using allocated slots vs total slots.
5. Determine total number of allocated slots for active users.
6. Plot time slot allocation and utilization using MATLAB graphs.
7. Display results in the command window for analysis..

Matlab Code :

```
clc; clear; close all;
```

```
FrameTime = 4.615; Slots = 8;
```

```
SlotDuration = FrameTime/Slots;
```

```
t = 0:SlotDuration:FrameTime;
```

```
figure
```

```
subplot(2,1,1)
```

```
stem(t,ones(size(t)),'filled')
```

```
title('TDMA Time Slot Structure')
```

```
xlabel('Time (ms)')
```

```
ylabel('Slot Level')
```

```
grid on
```

```
subplot(2,1,2)
```

```
plot(t,ones(size(t)),'-o','LineWidth',1.5)
```

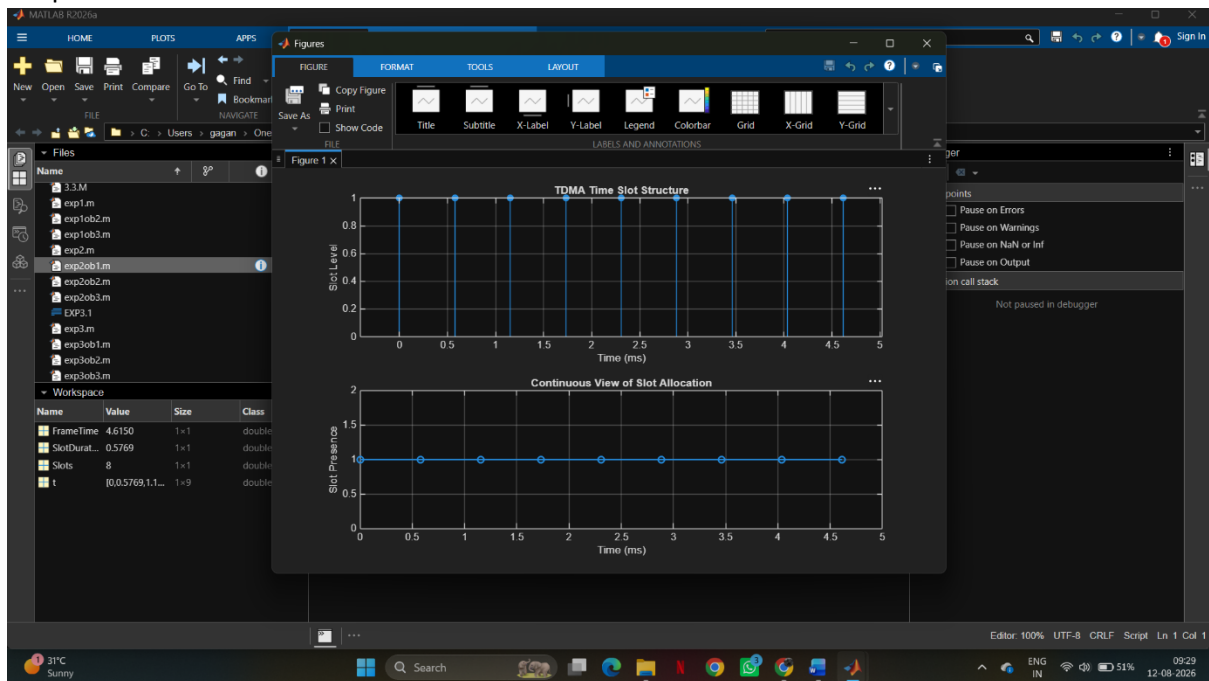
```
title('Continuous View of Slot Allocation')
```

```
xlabel('Time (ms)')
```

```
ylabel('Slot Presence')
```

```
grid on
```

Output:



Result:

1. The time slot duration in the GSM TDMA frame was successfully calculated, confirming equal division of the frame into fixed time intervals.
2. The frame utilization was effectively analyzed, showing the proportion of allocated and free slots in the TDMA system.
3. The slot allocation per user was successfully implemented, demonstrating that limited slots are assigned to users while excess users are blocked due to resource constraints.